Noetic Diffusion Theory: Neural State, Dynamical Accessibility, and the Geometry of Possible Futures

A multiscale framework for rhythmically and contextually controlled neural dynamics

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Abstract Noetic Diffusion Theory (NDT) proposes that neural-state organization should be characterized not only by where a system is located in state space, but by which transitions, perturbations, and future states remain dynamically accessible from that location. Earlier formulations emphasized rhythmic variance control, denoising on a learned potential, and representation in Meta–Noetic Phase Space (MNPS). The present framework generalizes that account into a layered dynamical theory: measured state geometry, local deterministic flow, local stochastic diffusion geometry, stability and reactivity, short-horizon reachability, basin commitment, finite-amplitude resilience, and candidate biological mechanisms that control accessibility. MNPS 3D and stratified MNPS 9D are retained as canonical measurement charts. NeuralManifoldDynamics is separated from NDT proper as one versioned empirical realization of those charts. Rhythmic Variance Control is retained as the minimal accessibility-control mechanism, not as the whole theory. Companion truth-known simulations qualify selected diffusion and Jacobian-derived summaries under restricted conditions; empirical work across sleep, altered states, clinical cohorts, and meso-to-macro recordings supplies bounded constraints rather than validation of the chain as a whole. NDT therefore predicts structured dissociations among state location, local stability, stochastic spread, accessible futures, basin commitment, and recovery from perturbation, rather than a single scalar of consciousness.

Author Summary

This paper rebuilds Noetic Diffusion Theory around a simple idea: a neural state is not fully described by where it sits. What matters scientifically is also what the system can still do from there — which directions it tends to move, how fluctuations spread, which nearby futures remain open, and what control processes change those possibilities. We keep a measurable state-space chart, but we treat it as an observation map rather than as the theory itself.

1. Scope and epistemic status

This article advances Noetic Diffusion Theory as a framework for neural dynamical organization associated with conscious states and cognition, not as a completed theory of why neural activity is accompanied by experience (Langell, 2025; Storm et al., 2024). The scientifically relevant object is a neural state's **dynamical accessibility**: the structured set of transitions, perturbations, destinations, and recoveries that remain available from the current condition, under deterministic, stochastic, contextual, and physiological constraints.

That object is distinct from several things with which it is easily confused. NDT is not a consciousness scalar. It is not identical to Meta–Noetic Phase Space, which is a measurement chart. It is not identical to NeuralManifoldDynamics, which is a versioned empirical implementation of selected charts and estimators. It does not claim that one thalamic circuit, one rhythm, or one diffusion parameter uniquely implements consciousness, nor that every quantity named below currently has a validated empirical estimator.

1.1. Evidence classes

Every major statement in this manuscript is intended to be readable as one of five classes:

1. **Established external result** — supported by peer-reviewed literature or other independent evidence outside this programme.

2. **Internal validated result** — supported by auditable companion code, tests, or frozen simulation/empirical contracts.
3. **Plausible interpretation** — compatible with present evidence but not directly demonstrated.
4. **Speculative extension** — a hypothesis, analogy, or forward model.
5. **Rejected or method-limited path** — tested and not supported under the stated assumptions.

Promotion rule

Success at one inferential level does not authorize a stronger claim at another. Correlation is not mechanism. Chart separation is not latent-state identification. Preserved latent ordering is not recovery of a local dynamical operator. Truth-known estimator recovery is not empirical biological validation.

The remainder of the article is organized by four layers that will be used as an editorial filter when companion results are imported: state geometry, local dynamical geometry, geometry of possible futures, and control of accessibility. MNPS belongs primarily to the first layer. Much of the recent programme — local Jacobians, diffusion tensors, reachability, destination geometry, and finite-amplitude questions — belongs to the latter three.

2. Introduction

Neural systems are often summarized by where they sit: a coordinate in a reduced state space, a complexity index, a spectral signature, a reported conscious level. Those summaries can be useful, and metastable state-space descriptions are a natural neighbourhood for the present work (Deco et al., 2017; Tognoli & Kelso, 2014). They are incomplete as an account of regime. Two systems can occupy similar locations and still differ in how nearby perturbations grow, how stochastic variation spreads, which future volumes remain reachable, which basins are favoured, and whether a finite kick can dislodge the current regime.

Earlier NDT made that incompleteness only partly explicit. Its identity was rhythmic reconstruction: a Meta–Noetic Diffusion Model (MNDM) as guided stochastic flow, Rhythmic Variance Control (RVC) as a schedule on scalar noise amplitude, and MNPS as the Stage-1 chart in which reconstruction should become visible (Langell, 2025). The implied chain was approximately

rhythmic control → variance scheduling → exploration or denoising → MNPS geometry.

That chain remains a legitimate special case. It is no longer an adequate centre of gravity. The programme has since been forced, by both positive and negative companion results, to treat local flow, diffusion geometry, and accessible futures as first-class theoretical objects, and to treat measurement charts as operational rather than ontological.

The revised chain is:

state → local dynamical law → possible futures → state-dependent control of those futures.

The core claim is therefore not merely that conscious-state correlates occupy characteristic regions of neural geometry. It is that the functional significance of a neural state depends partly on the structured set of transitions available from it.

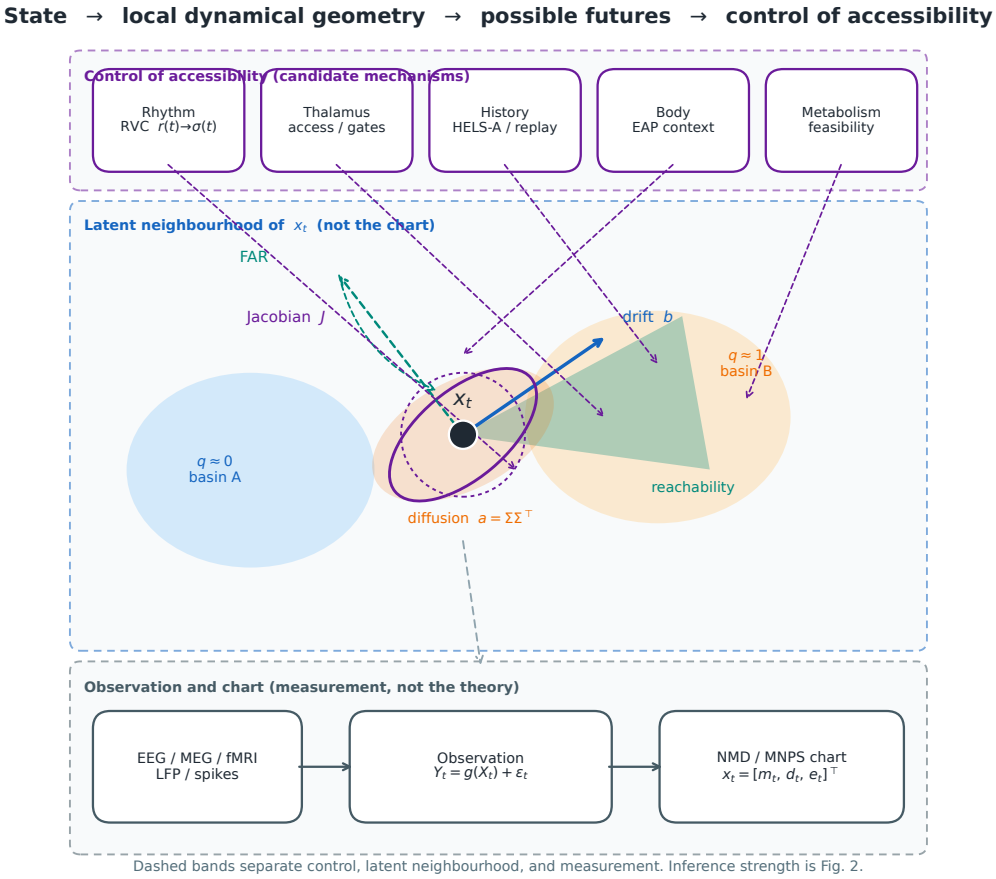


Figure 1: The NDT 1.0 architecture. A measured point x_t is surrounded by local dynamical objects (drift, diffusion, Jacobian deformation, reachability, committor basins, finite-amplitude perturbation). Candidate controls act from above on different parts of that neighbourhood. EEG and related recordings enter only through an observation map and a versioned chart. Dashed bands separate control, latent dynamics, and measurement.

Core question

How should neural-state organization associated with cognition and conscious regimes be characterized if location and complexity are insufficient — that is, if the scientifically relevant object includes local flow, stochastic spread, reachable futures, basin commitment, and the processes that reshape them — without mistaking a measurement chart for the latent process or a correlation for a dynamical operator?

2.1. Three questions

The framework is organized around three questions that later sections develop in turn.

Where is the system? This is the measurement-chart question. A low-dimensional representation

$$x_t = [m_t, d_t, e_t]^\top \tag{1}$$

or a stratified nine-dimensional refinement, reports an operational location. It answers a **state** question.

What can happen from here? This is NDT proper. It includes deterministic drift, stochastic diffusion, transient amplification, short-horizon reachability, basin commitment, and finite-amplitude resilience. These objects answer different dynamical questions and are not interchangeable summaries of one another.

What changes what can happen? This is the control layer: rhythms, candidate thalamic access, attention and context, history and replay, bodily state, metabolic feasibility, and — as an explicitly speculative micro substrate — slow structural accessibility at dendritic scale. Control may act on drift, diffusion, effective geometry, landscape, or transition gates, not only on a scalar noise amplitude.

2.2. Noetic accessibility

Noetic accessibility, as used here, is a dynamical-systems concept: the structured, state-dependent set of dynamically attainable neural transitions and destinations under the current deterministic, stochastic, contextual, and physiological constraints. The adjective "noetic" retains its earlier sense of knowledge-bearing or intelligible structure (Langell, 2025). It does not mean conscious access in the global-workspace sense (Dehaene & Changeux, 2011). Reachable volume is not workspace ignition. Chart-level accessibility is not a report.

3. The NDT scale and inference ladder

3.1. Latent process, observation, and chart

Let X_t denote a latent neural dynamical state. Its exact dimensionality and biological substrate need not be known for the theory to be well posed as a modelling class. Measurements arise through an observation map

$$Y_t = g(X_t) + \varepsilon_t, \tag{2}$$

where Y_t may be spikes, local field potentials, EEG, MEG, fMRI, or physiological channels, and ε_t is observation noise. A subsequent feature and chart construction produces an operational representation

$$Y_t \to z_t \to x_t. \tag{3}$$

The chart x_t is a versioned summary of the measurement. It is not automatically the latent manifold. Different modalities, preprocessing decisions, neighbourhood definitions, and projections yield different charts, and their uncertainty is inherited by every later estimate of drift, Jacobian, diffusion, or reachability.

3.2. Four levels of recoverability

Companion truth-known work on observability and local-operator recovery motivates an operational ladder that governs claim strength throughout this article:

1. **Association.** Does an observable covary with another measured or latent quantity under stated nulls?
2. **Mechanism discrimination.** Can observations distinguish alternative mechanisms that produce similar task-level outputs?
3. **Latent-order preservation.** Does the measured representation preserve a predefined ordering of a latent summary?
4. **Dynamical-operator identification.** Can the representation recover a local operator that correctly predicts responses to perturbation or short-horizon evolution?

These levels are not a general nonlinear-observability theorem. They are a claim ladder. The ladder is not monotonic: a chart can discriminate mechanisms while failing to preserve order or to identify an operator, and an observation map can destroy a relation that was recoverable in a cleaner representation.

A companion meso-to-macro analysis of simultaneous Neuropixels and surface EEG illustrates the rule rather than validating NDT geometry. Bounded associations between mesoscopic population summaries and EEG variance or spectral entropy survive temporal and statistical nulls at the level of association. The same EEG-only local-linear chart does not support Jacobian or reachability inference: the residual-variance fraction of the fitted local operator remains too large for operator identification. Cross-scale contact is therefore a Level-1 constraint. It is not common latent-state identity, unique NDT geometry, or a validated chart Jacobian.

Consequence

MNPS separations can be meaningful at Levels 1–3. Claims about Jacobians, diffusion tensors, reachability, committors, or finite-amplitude resilience require stronger validity conditions and fail closed when those conditions are not met.

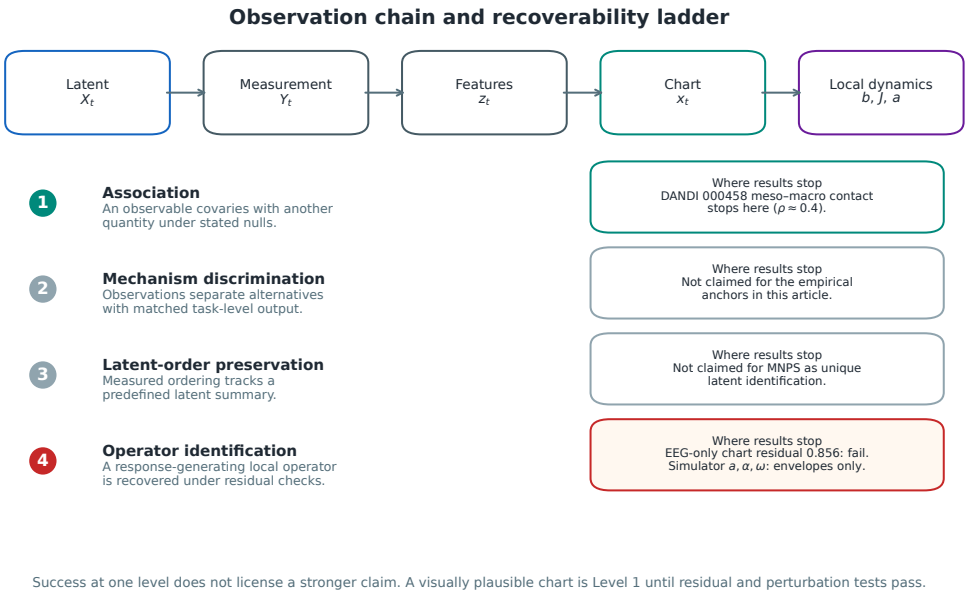


Figure 2: Observation chain and recoverability ladder. Companion meso-to-macro contact on DANDI 000458 is a Level-1 constraint. The same EEG-only local-linear chart fails operator identification. Truth-known recovery of a, α, and ω remains envelope-limited.

4. Core NDT dynamics

4.1. Canonical stochastic formulation

The working process class of NDT 1.0 is a controlled diffusion, not a pure gradient flow. Let c_t denote a contextual or control state. The latent dynamics are written

$$dX_t = b(X_t, t \mid c_t)dt + \Sigma(X_t, t \mid c_t)dW_t, \tag{4}$$

where b is local deterministic drift, Σ determines the amplitude and orientation of stochastic forcing, and W_t is Brownian motion. Equation 4 is a modelling class, not a fitted neural SDE. Data constrain a local generator more directly than they identify a unique decomposition into metric, potential, and noise, and the representation remains subject to chart changes, time reparametrization, and additive constants in any potential.

4.2. Diffusion geometry

Diffusion in this theory is not a scalar noise amplitude. Define the local diffusion tensor

$$a(x, t) = \Sigma(x, t)\Sigma(x, t)^\top. \tag{5}$$

The tensor a determines the magnitude, effective dimensionality, anisotropy, and preferred directions of stochastic spread. Two states with identical mean drift may differ profoundly in what fluctuations permit. Scalar $\sigma(t)$ is the isotropic special case.

4.3. Potential-driven and non-conservative drift

The original landscape intuition is preserved as a structured component of drift, not as an assumption that every neural trajectory is gradient descent:

$$b(x, t \mid c) = -G(x, t \mid c)\nabla\Phi(x, t \mid c) + u_{\mathrm{nc}}(x, t \mid c). \tag{6}$$

Here Φ is a state-dependent potential, G is an effective metric or mobility geometry, and u_{nc} collects non-conservative contributions: rotational flow, recurrent cycling, asymmetric transport, non-normal transient dynamics, and context-conditioned routing. Pure potential descent cannot naturally represent reactivity with decaying modes, rotational structure, or asymmetric access. The split in Eq. 6 keeps the landscape reading available while making those phenomena first-class.

4.4. Rhythmic variance control as the minimal special case

Rhythmic Variance Control remains the simplest accessibility-control mechanism (Langell, 2025). In the original scalar form,

$$\sigma(t) = \sigma_{\min} + \sigma_0(1 - r(t)), \tag{7}$$

with $r(t) \in [0, 1]$ an operational rhythmic-control proxy: high r lowers noise amplitude. The present framework retains Eq. 7 as a special case of

$$r(t), c(t) \to \{b_t, \Sigma_t, G_t, \text{transition gates}\}. \tag{8}$$

Rhythm may still schedule exploration versus stabilization. It is no longer asked to carry the whole theory. Stronger specificity claims are treated as control-layer hypotheses in Section 9, not as denoising proofs.

5. Measured state: MNPS and the NeuralManifoldDynamics boundary

5.1. Canonical and stratified charts

The canonical three-dimensional observation chart remains Eq. 1. Operational families are unchanged in spirit: m indexes mobility, metastability, or rhythmic coordination; d indexes deviation from an integration–segregation balance; e indexes entropy or complexity pressure. These are chart semantics, not unique biological axes.

A stratified nine-dimensional chart

$$x_t^{(9)} = [m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m]^\top \tag{9}$$

is retained as higher-resolution measurement, with recomposition into the three families through a release-fixed projection $x_t = P_{\text{fixed}} x_t^{(9)}$. The rationale is to reduce dimensional masking and to expose compensatory changes hidden in 3D. P_{fixed} is an auditable operational contract, not a claim of unique ontology. Downstream local-dynamics claims inherit that contract.

5.2. MNPS is not NDT

MNPS provides coordinates in which some NDT-related quantities can be estimated. It does not exhaust NDT. The theory remains meaningful if another valid chart replaces MNPS, if some dynamical objects are more identifiable than the coordinates themselves, or if modality-specific charts differ. Identifying the latent state X_t with (m, d, e) is rejected as a homology.

5.3. NeuralManifoldDynamics as a measurement contract

NeuralManifoldDynamics (NMD) is a versioned empirical measurement contract that implements selected MNPS and local-dynamics quantities. MNPS is an observation chart through which some NDT quantities can be estimated, not the theory itself.

NDT owns latent interpretation, the objects b, a, J, reachability, committor, and finite-amplitude questions, predicted dissociations, and mechanism hypotheses. NMD owns windows, transforms, coverage, feature-to-chart maps, the fixed projection P_{fixed}, HDF5 export semantics, and particular estimators where implemented.

In current NMD releases the three- and nine-dimensional charts are core serialized objects. A chart Jacobian J_{hat} estimated in MNPS is an **MNJ** — a Jacobian in that chart — not a generic $J(X_t)$. Derived stability and reactivity summaries, when emitted, are per-recording mathematical measurements with provenance; they are not cohort statistics and not biological claims. Stochastic reachability is gated: derivative-residual covariance is not an admissible one-step noise tensor without an explicit conversion model, and production summarization does not currently emit computed reachability without that contract. Committor and finite-amplitude-resilience families, where present at all, are opt-in and refuse computation without explicit regime sets or perturbation protocols. Companion truth-known recovery of diffusion tensors and Jacobian summaries on linear SDEs does not, by itself, license literal ingest of the same quantities through the current NMD pipeline.

5.4. Measurement inheritance rule

Any downstream inference inherits the validity limits of (i) raw measurement, (ii) preprocessing, (iii) feature construction, (iv) chart definition, and (v) local estimator. A plausible embedding does not cancel those limits.

6. Local dynamical geometry

Given a representation in which a drift b is defined, the local Jacobian is

$$J(x, t) = \frac{\partial b}{\partial x}. \tag{10}$$

The generic object is this J. The term MNJ is reserved for a Jacobian estimated in an MNPS chart. J asks how nearby perturbations change under the local flow. It does not ask where the system sits, how noise spreads, or which basin is favoured.

Two derived summaries are conceptually distinct and must remain so. The **spectral abscissa**

$$\alpha(J) = \max_i \Re[\lambda_i(J)] \tag{11}$$

is the dominant modal growth or decay rate. The **numerical abscissa**

$$\omega(J) = \lambda_{\max}\left(\frac{J + J^\top}{2}\right) \tag{12}$$

is the maximal instantaneous growth rate. A system can satisfy $\alpha < 0$ while $\omega > 0$: every eigenmode decays, yet some perturbations grow at short times. The reactivity gap $\Delta_{\text{react}} = \omega - \alpha$ is a derived, non-normality-sensitive summary of that separation. It is not a complete measure of non-normality. The antisymmetric part $\Omega = \frac{J - J^\top}{2}$ supplies a rotational component distinct from reactivity.

Finite-time amplification, for example a peak gain $G_{\text{peak}}(T)$ of a local propagator over a horizon T, is a different object again. Companion truth-known work recovers α, ω, and Δ_{react} on declared constant-linear systems under restricted envelopes, and shows that they are not interchangeable. Instantaneous Jacobian recovery does not automatically license ordered finite-time promotion, and simulator success does not automatically transfer to literal NMD ingest. Those boundaries stay in force here: "stable" is not a sufficient dynamical phenotype.

7. Local diffusion geometry

Equation 5 already makes diffusion a tensor. The questions it answers are orthogonal to those answered by J. Natural summaries include total diffusion $D_{\text{total}} = \operatorname{tr}(a)$, an effective diffusion dimensionality, a concentration or anisotropy index, and orientation-specific alignments relative to a declared probe direction.

Two states can share a similar J and differ in a, or share a similar a and differ in J. That dissociation is part of what "diffusion" is supposed to mean in NDT. Companion truth-known recovery of a on linear SDEs supports the object as a mathematical measurement under inspectable envelopes (support, timestep, bandwidth, observation noise). Gauge matters: orthogonal rotations leave tensor summaries invariant to numerical tolerance, whereas nonuniform rescaling does not. Recovered a is therefore chart-dependent unless a chart is declared. Empirical interpretation through NMD remains a separate translation problem; the current pipeline treats diffusion geometry as an opt-in family with explicit qualification gates, not as a silent by-product of Jacobian residuals.

8. Geometry of possible futures

Local linearization does not exhaust accessibility. This section collects objects that ask what can happen over a finite horizon, toward which destination, and after a perturbation that is no longer infinitesimal.

8.1. Reachability

Short-horizon reachability asks which nearby future states remain dynamically available. A local stochastic linearization

$$x_{t+1} = A_t x_t + b_t + \varepsilon_t, \quad \varepsilon_t \sim (0, Q_t) \tag{13}$$

induces the covariance recursion

$$\Sigma_t^{(h+1)} = A_t \Sigma_t^{(h)} A_t^\top + Q_t, \tag{14}$$

with $\Sigma_t^{(0)} = 0$. Eigenvalues of $\Sigma_t^{(h)}$ yield volume, anisotropy, and effective-dimension summaries of an ellipsoidal reachable set.

Reachability is not occupancy, entropy, traversability, controllability, or consciousness. Large Q_t can inflate cones without richer controllable geometry. Horizon and neighbourhood leakage are validity conditions. Companion simulator work can calibrate aggregate covariance propagation while failing tests of memory-specific or target-private directional spread; calibration is not destination geometry. On the measurement side, NMD does not currently emit computed stochastic reachability without an admissible one-step residual contract. Empirical claims that invoke reachable futures therefore require their own estimator audit; they cannot be read off chart location.

8.2. Committor geometry

For competing destinations A and B, the committor

$$q(x) = P_x(\tau_B < \tau_A) \tag{15}$$

is the probability of hitting B before A. It is destination geometry, not local expansion and not reachable volume. The object is theoretically central. Its empirical and even finite-sample status is severely bounded.

One-dimensional local-law recovery can be qualified under restricted contracts; two-dimensional finite-sample recovery has not been licensed under the estimator families so far tested, and a product-level committor export in NMD is not authorized by that stop. Stage labels are not committor truth. The flagship therefore promotes the **question** and refuses silent computation.

8.3. Attractors and finite-amplitude resilience

Attractor and basin structure — locations, depth, boundaries, corridors, metastability, escape — organize longer-run trajectories (Deco et al., 2017; Tognoli & Kelso, 2014). Visual clustering is not evidence of an attractor. Finite-amplitude resilience (FAR) asks how large a perturbation the system can tolerate before leaving its current dynamical regime. Local stability does not imply FAR; Jacobian or finite-time tangent gain does not substitute for an observed perturbation protocol. NMD retrospective ingest correctly reports FAR as not testable without such a protocol.

8.4. Orthogonal dynamical questions

The point of this hierarchy is not to brand a new metric for each row. It is to keep the questions distinct.

Dynamical question	**Object**
Where is the system?	chart state x
Where is it locally moving?	drift b
Do infinitesimal modes grow?	spectral abscissa α
Can perturbations initially amplify?	numerical abscissa ω
How does random variation spread?	diffusion tensor a
Which nearby futures are available?	reachability
Which destination is favoured?	committor q
Which basin organizes the trajectory?	attractor geometry
Can a finite perturbation dislodge it?	FAR

Table 1: Orthogonal questions in NDT. No single local measure answers the others. Empirical availability of each object is a separate contract, not a consequence of having a chart.

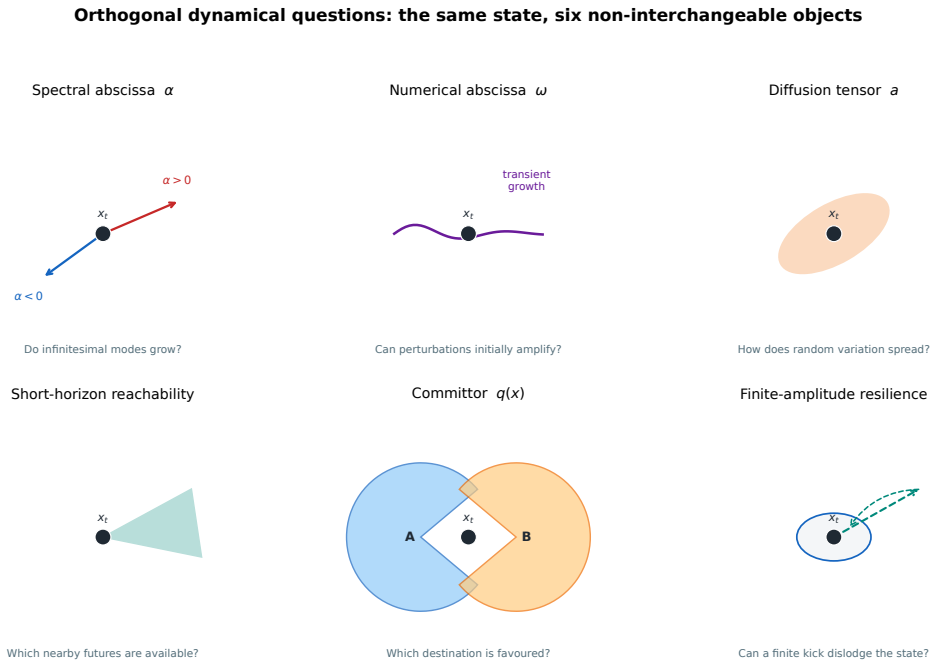


Figure 3: Orthogonal dynamical questions. The same state x_t supports six non-interchangeable objects. No single local measure answers the others.

Formal neighbours — network control, predictive processing, metastability, and the major consciousness theories — are compared in Section 11. The present emphasis remains measured dynamical accessibility rather than control energy, free-energy identity, workspace ignition, or integrated information.

9. Biological control of accessibility

Equation 4 is incomplete until one says what c_t is allowed to change. In NDT 1.0, control may act on drift, diffusion, mobility geometry, landscape, or transition gates — not only on a scalar noise amplitude (Eq. 8). Different candidate mechanisms occupy different slots in that list. They are not interchangeable implementations of one controller, and a failure in one slot does not falsify accessibility control as a modelling class.

Candidate	**Intended action**	**Present ceiling**
RVC	scalar $\sigma(t)$	minimal special case; event-gated write specificity not supported
Thalamic access	directional A_{sig}, A_{noise} on frozen Φ	synthetic operator existence; biology is association, not operator ID
HELS-A	context operator A_H on frozen Φ	synthetic access without landscape rewrite; empirical scoring not licensed
HELS (legacy)	$\partial\Phi/\partial t$ from history	landscape problem remains open; infrastructure is not writing
Embodiment / metabolism	context c_t^{body}; feasibility q^{met}	conditioning and constraint; not consciousness coordinates
Dendritic M_b	slow branch access $A^f A^s$	simulator discriminator; speculative substrate, not MNPS identity

Table 2: Control of accessibility. Each row changes a different part of the local dynamical object. None is licensed as **the** NDT mechanism.

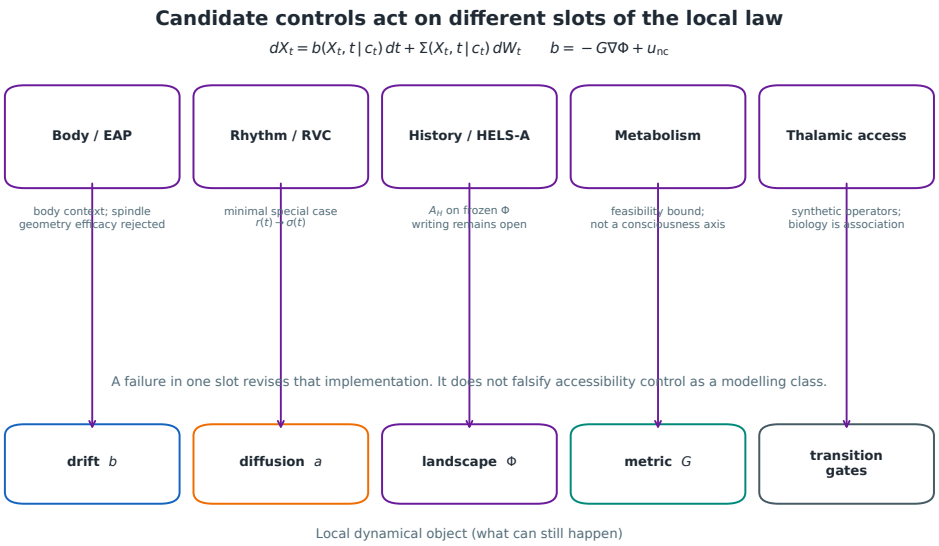


Figure 4: Accessibility control. Each candidate mechanism is drawn as acting on a distinct slot of the local law. A failure in one slot revises that implementation; it does not falsify the modelling class.

9.1. Rhythmic variance control

Equation 7 remains the simplest NDT controller: a rhythmic proxy $r(t)$ schedules isotropic noise amplitude and thereby the exploration–stabilization trade-off (Langell, 2025). That is a claim about **one** entry in $\{b, \Sigma, G, \Phi, \text{gates}\}$. It is not a claim that physiologically discrete events write memory, nor that any event-locked chart reconfiguration is denoising.

Companion simulator tests of stronger controllers — periodic or event-triggered **write-gate** timing against energy-matched and shuffled-timing controls — have not supported a general rhythmic-necessity claim. Those negatives constrain P1-style specificity hypotheses. They do not retract scalar RVC as a modelling special case, and they do not license reading sleep spindles as proven variance clamps. Event-locking of a measurement

chart, if replicable, is a state-geometry constraint. Causal denoising remains a separate, currently unsupported, promotion.

9.2. Thalamic access as candidate architecture

The obsolete anatomical slogan was that the thalamic reticular nucleus implements a variance clamp. The present hypothesis is weaker anatomically and stronger computationally: thalamocortical and thalamoreticular circuitry is a **candidate access-control architecture**. It may gate channels, change effective variance or population dimensionality, rotate diffusion, shape local transition geometry, or route context — none of which is required to be scalar $\sigma(t)$.

A companion two-operator specialization makes that distinction executable on a frozen landscape,

$$dX_t = A_{\text{sig}} b_0(X_t)dt + A_{\text{noise}}\Sigma_0 dW_t, \tag{16}$$

with Φ held fixed. Under that contract, directional A_{noise} can rotate short-horizon dispersion at matched total diffusion power; non-normal A_{sig} can produce $\alpha < 0 < \omega$ while a scalar contraction cannot; and finite-time gain and Jacobian–noise interaction can split from scalar controls. Those are **operator-existence** results on a declared 2-D potential. They are not thalamic physiology, not MNPS, and not consciousness. Native-domain finite-amplitude recovery has not been licensed as a confirmatory result.

Empirical TRN perturbation work currently reaches a lower rung. In one auditory TRN-to-relay preparation, PV-selective photoinhibition is associated with evoked participation-ratio contraction; a full TRN variance-gate hypothesis, including a scalar Fano clamp, is not validated. On the same resource, both PV- and SST-family photoinhibition associate with non-scalar change in residual covariance **shape** after prestimulus cleaning — a bounded contrast against scalar rescaling, not identification of A_{noise} or of a chart Jacobian. Stronger repertoire-gating and prestimulus-controlled shape claims on a separate optogenetic resource have been rejected. Direct mapping onto NDT operators remains open. NDT has not found the thalamic mechanism.

9.3. History: landscape writing versus access without rewriting Φ

The landscape problem is independent of thalamic routing. Equation 6 can specify motion given Φ. It does not say where Φ came from. A history-conditioned landscape

$$\Phi(x, t \mid H_t) \tag{17}$$

is the natural object for later transition probabilities, basin depth, corridor availability, and hysteresis. The original Hippocampal–Entorhinal Landscape System (HELS) programme asked whether replay and related hippocampal dynamics **write** that landscape. Current infrastructure and decoder-feasibility work on public hippocampal archives do not constitute evidence of landscape writing. That HELS question remains a speculative mechanistic subsystem.

HELS-A is a different question, and collapsing the two names is a category error. HELS-A asks whether a context-conditioned access operator can change which routes remain available **while Φ is held fixed**:

$$dX_t = -A_H \nabla\Phi(X_t)dt + \sigma dW_t. \tag{18}$$

Companion truth-known development gates support that split on declared geometries: directional A_H can saturate first-passage toward a context-matched target where speed-matched scalar gain does not, and an unsaturated double-well committor can be biased by context, reversed by the wrong context, and collapsed by shuffle. A neighbourhood extension failed its coded robustness gate and is not treated as a pass. Official confirmatory seeds remain unconsumed. The licensed statement is therefore narrow. A fixed stored potential does not uniquely determine route accessibility: Φ fixed and X_0 fixed do not imply $q(X_0)$ fixed when A_H changes. That is not a hippocampal, thalamic, memory, or MNPS result. It also does not answer "where does Φ come from?" HELS-A shows that the landscape problem and the access problem can come apart.

The biological HELS-A1 gap is still open. A simultaneous CA1–prefrontal candidate archive can be structurally eligible at the header level — region labels, trial clocks, and behavioural U_t sufficient to **attempt** a later design — without that eligibility being a mechanism. Fabricated-estimator qualification for a state-matched incremental test of residual hippocampal information about future cortical direction has not passed family false-positive control. Conditional predictive information is not causal access. Incomplete observation of current state and of task/context state are identifiability ceilings, not evidence that CA1 implements A_H. Spike-time scoring is closed until an estimator is independently qualified. Header yield is not a hippocampal result.

A separate empirical observation — that true recent history can improve linear prediction of post-stimulation measured EEG beyond the current measured state on one meso-to-macro contract — is a P6-adjacent **measured-state** constraint. It is not HELS landscape writing and not HELS-A1.

9.4. Embodied state and metabolic feasibility

Bodily signals are treated as context variables that may condition accessibility, not as coordinates of consciousness. Fast candidates include cardiac and respiratory phase, autonomic state, interoception, and heartbeat-evoked responses. Companion work can support bounded autonomic–chart co-modulation without identifying causal body-to-manifold gating. Any such coupling belongs in c_t^{body}, not in (m, d, e) by default.

Slower physiology is a feasibility boundary rather than another rhythmic anchor. Let $q^{\mathrm{met}(t)}$ denote metabolic adequacy. A trajectory may be geometrically available in the abstract model yet physiologically unavailable because the substrate cannot support it. Metabolism is not a consciousness scalar, not an MNPS axis, and not equivalent to arousal. Equations that fold q^{met} into an effective rhythmic schedule remain speculative extensions for later tests.

9.5. Slow structural accessibility at dendritic scale

Accessibility control is not confined a priori to thalamic or EEG scales. A companion branch-resolved memory simulator separates fast dendritic access from a slower phenomenological bias M_b, with effective access

$$A_{b(t)} = A_b^f(t) A_b^s\big(M_{b(t)}\big). \tag{19}$$

Replay-dependent updating of M_b can reproduce a joint profile of overlap-branch strengthening, linking gain, perturbation-sensitive loss, and targeted rescue that tested comparators do not reproduce, inside a bounded write–replay and non-hub motif regime. M_b is not a unique cytoskeletal memory code. Open-data bridges that do not observe M_b are not primary evidence. Exporting $\{x_b, M_b, ...\}$ into MNPS, MNJ, or reachability remains a speculative forward model. The flagship import is conceptual: the same logical move — control changing futures without rewriting content — can be stated at the dendrite, and the micro–meso–macro chain is still mostly dashed.

10. Empirical constraints

No companion result validates the NDT chain end to end. The role of evidence here is to constrain layers, not to define the theory. Three anchors enter the main text because they answer different questions in the hierarchy of Table 1. Everything else is summarized as structured dissociation rather than narrated dataset by dataset.

Companion truth-known simulations, already used to bound a, α, and ω in Section 6 and Section 7, belong in this map as **methodological qualification**: they ask whether nominated mathematical objects recover known latent targets under inspectable envelopes. They do not identify simulator coordinates with MNPS and they do not replace empirical anchors.

10.1. Anchor 1: spindle-locked state reconfiguration

The first question is whether a physiologically discrete rhythmic event coincides with local chart reconfiguration — not whether spindles prove denoising, memory writing, or a thalamic variance clamp.

A companion three-dataset analysis of N2 spindles on a frozen stratified MNPS export finds a replicable within-event rise in the network-deviation coordinate d_n when the spindle window is compared with a far pre-event baseline (ANPHY high-density EEG, $N = 27$; BOAS / OpenNeuro ds005555, $N = 128$) (López-Larraz et al., 2025; Wei et al., 2024). Effect sizes on that timing contrast are modest (d_n shifts of order 0.11–0.17 after FDR). A third overnight polysomnography sample is underpowered for the same timing estimand and is treated as corroboration, not as an independent timing replication. Jacobian, diffusion-tensor, and reachability endpoints were not the primary objects.

Matched-control specificity is incomplete. Several family endpoints are dominated by baseline differences rather than by the event increment, and the event-versus-matched-control sign for d_n conflicts across datasets for that reason. Autonomic composites do not moderate the geometry effect under the primary protocol. Sigma-band power residualization was not part of the frozen schema.

Anchor 1 ceiling

Discrete spindle events coincide with a replicable, timing-locked rise in stratified d_n. That is chart-level state reconfiguration at a rhythm-defined micro-event (recoverability Level 1). It is not causal denoising, not a pure spindle-versus-matched-N2 mechanism, not a reachability change, and not TRN identification.

10.2. Anchor 2: recovery-linked geometry after cardiac-arrest coma

The second question is whether a clinically important state difference appears more clearly in accessible futures and local flow than in static location alone.

A companion pooled analysis of early post-arrest EEG from I-CARE PhysioNet 2.1 evaluates MNPS spread, local Jacobian deformation, and short-horizon reachability jointly in two independently processed, non-overlapping cohorts (301 analyzable recordings; 281 in the CPC 1 versus CPC 5 contrast; window 0–12 h after the EEG clock, not the ROSC clock) (Amorim et al., 2023). All five prespecified reachability summaries survive BH-FDR in the pooled sample, led by persistence and log-det volume proxies (absolute Cohen d near 0.9). Local Jacobian summaries are directionally aligned — lower expansion and deformation in later-good-outcome recordings — but heterogeneous in magnitude across cohorts. Static MNPS spread is lower in better-outcome patients and is restored as a descriptive anchor, not as the lead. The leading reachability separator is already detectable in the first analyzable hour; a subject-wise circular-shift null does not support a claim about which hour is uniquely strongest. Adding the geometry layer to a feature-limited clinical-plus-qEEG baseline raises internal AUROC from 0.599 to 0.662, a bounded increment that is not multimodal bedside prognostication.

Because withdrawal-of-life-sustaining-therapy, time-to-death, and sedation annotations were unavailable, the contrast is recovery-linked geometry, not a pure viability assay and not a consciousness meter.

Anchor 2 ceiling

Early post-arrest EEG contains layered, recovery-linked geometry: short-horizon reachability and local deformation carry information that static severity/location does not exhaust. These are empirical NMD-chart estimators. Residual quality and perturbation-response were not the study design, so the result motivates the accessible-futures layer. It does not constitute truth-known dynamical-operator identification.

10.3. Anchor 3: meso-to-macro contact without operator recovery

The third question is methodological and was already used as a worked example in Section 3. Can mesoscopic population quantities predict macroscopic EEG features under nulls, and does that contact license a local operator?

A companion analysis of simultaneous Neuropixels and surface EEG during cortical electrical stimulation, continuing from wakefulness into isoflurane (Claar et al., 2023), reports bounded Level-1 associations: post-artifact LFP-variance proxies and quality-filtered spike rates covary with EEG variance and spectral entropy at median subject correlations near 0.4, surviving sign-flip and circular time-shift nulls. The same EEG-only local-linear chart yields a median residual-variance fraction of 0.856. Jacobian and reachability inference on that chart are therefore rejected.

The pair is the epistemic spine of the empirical map. Cross-scale contact is real at association. It is not common latent-state identity, unique NDT geometry, causal direction, TRN identification, or a validated MNJ. A visually plausible chart plus a correlation does not climb the recoverability ladder.

Anchor 3 ceiling

Meso and macro observables can covary under stimulation-aligned windows after temporal and statistical nulls (Level 1). The tested EEG chart does not recover a local flow, Jacobian, or reachability operator (Level 4 fail).

10.4. Structured dissociations, not a universal signature

NDT predicts that different conditions move different layers. Companion clinical and task analyses are therefore kept as one constraint matrix rather than as additional anchors.

Setting	What moves	What does not	Ceiling
Depressive-symptom rest	reachability; 9D rotational structure	coarse 3D volume as lead	symptom-burden geometry, one young cohort
First-episode psychosis	medication-stratified chart and trajectory splits	one FEP signature	bounded; not a universal deformation
Gambling vs cognitive control	local Jacobian / flow geometry	MNPS location; matched EDA	state-versus-flow dissociation
Hypoglycaemic clamp	3D/9D state; slow-rhythm increase	confirmatory reachability collapse	feasibility constraint, not q^{met} identification
Infant age	chart coordinates and classical qEEG	FDR-supported local-flow change	development as state, not futures

Table 3: Selected dissociations. NDT is constrained by which layer moves, not by a single consciousness-shaped manifold shift. Full companion reports remain outside this article.

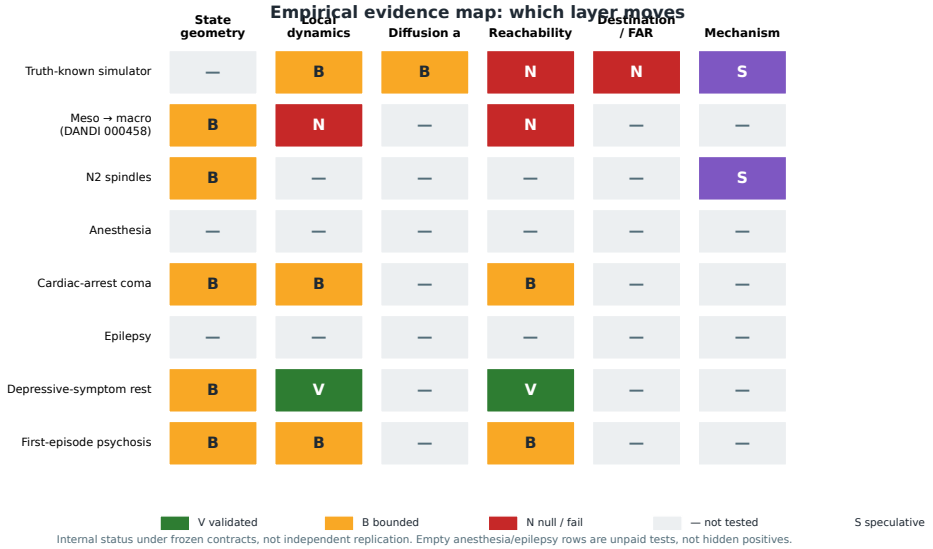


Figure 5: Evidence map for the flagship. Cells are internal status under frozen contracts, not independent replication. V = validated; B = bounded; N = null or fail; dash = not tested; S = speculative. Empty anesthesia and epilepsy rows are unpaid tests, not hidden positives.

The pattern is the result. Reduced dynamical capacity is not itself reduced consciousness. Sleep, anesthesia, seizure, coma, and affective or psychotic regimes may all alter capacity-like summaries while differing in whether the change is mainly location, local deformation, stochastic spread, reachable volume, or canalized lock-in.

11. Neighbouring frameworks

NDT does not compete with constitutive or access theories of consciousness on their own terms. Its defensible claim is narrower: it is a framework for dynamical accessibility and a versioned measurement programme on which predictions from several of those theories can be expressed and, in places, cross-checked (Storm et al., 2024). The comparison that follows is therefore a map of partly different questions, not a contest for a single title. A longer thematic treatment of the same neighbourhood exists as a companion essay; this section imports only the bridges and non-equivalences that constrain NDT 1.0.

Positioning

IIT proposes what consciousness is claimed to be. GNWT proposes when content becomes globally available. NDT proposes how neural trajectories are locally generated, spread, gated, and made more or less reachable. Those are not interchangeable answers to one question.

11.1. Different questions, not rival scalars

"What does consciousness correspond to?" is not a single question. Neighbouring frameworks answer at least six different ones: whether experience is present (level); what is experienced (content); whether information is available to report and action (access); whether the system represents itself as being in a state (meta-awareness); whether experience is embodied; and what range of neural transitions remains available (dynamical capacity). NDT addresses the last of these. It does not currently identify NCC proper, and its charts are closest to enabling state dynamics — NCC prerequisites and the dynamical context around experience — rather than to report, confidence, or phenomenal identity.

Theories can therefore be nested as explanatory layers without being jointly validated (Storm et al., 2024). Predictive inference can supply candidate content (Clark, 2013; Friston, 2010). Recurrent processing can stabilize sensory representations (Lamme, 2006; 2010). A workspace can broadcast selected content (Baars, 1988; Dehaene & Changeux, 2011; Mashour et al., 2020). Higher-order or attention-schema systems can represent aspects of that access (Brown et al., 2019; Lau & Rosenthal, 2011; Webb & Graziano, 2015). Embodied inference can bias the background state (Seth, 2013). Dynamical geometry can describe the trajectory through which those processes unfold. Structural network control can constrain which transitions are cheap or costly (Gu et al., 2015; Tang & Bassett, 2018). Compatibility of layers is not evidence that every theory is true.

11.2. Closest family: landscapes, metastability, and control

NDT's nearest relatives are dynamical-landscape, metastability, and network-control accounts (Deco et al., 2011; 2017; Gu et al., 2015; Kelso, 2012; Tognoli & Kelso, 2014). They already deny that regional activity magnitude is a sufficient description. NDT inherits that denial and adds five commitments that those families do not jointly require:

1. **Stochastic diffusion geometry** as a first-class local object $a(x, t)$, not only noise around an attractor.
2. **Accessibility decomposition**: location, local flow, stochastic spread, reachability, commitment, and finite-amplitude resilience answer different questions (Table 1).
3. **Explicit measurement contracts**: MNPS and NeuralManifoldDynamics are operational charts and estimators, not unique latent identification.
4. **State-dependent control**, of which rhythmic variance scheduling is the minimal special case, not the whole theory.
5. **Recoverability discipline**: association does not license a local operator (Section 3).

Network control theory shares the concern with possible transitions. NDT does not currently assume an explicit input matrix B or minimize control energy. Its reachability objects are empirical, chart-level, and short-horizon; they are not Gramian controllability of a structural connectome. Metastability shares the concern with a regime between lock-in and fragmentation. MNPS m is an operational mobility/coordination proxy, not a Kuramoto order parameter computed from pairwise phase relations. Landscape models share potential descent. In NDT that descent is a structured component of drift, not the general law: non-conservative flow, non-normal transients, and anisotropic diffusion remain in u_{nc} and a.

The claim "conscious states have geometry" is therefore not distinctive. The testable NDT increment is that neural variability and transition structure are actively controlled, and that the resulting local flow, diffusion, and accessible futures can contain information not recoverable from location, power, or generic complexity alone.

11.3. Constitutive, access, and inference theories

Framework	Possible bridge	Non-equivalence
PP / FEP	precision; landscape descent; state estimation	fitted accessibility is not a generative model with explicit priors
IIT	both reject activity magnitude as sufficient	NDT does not identify consciousness with Φ or intrinsic causal structure
GNWT	access geometry may enable or disable global availability	dynamical reachability is not workspace ignition
RPT	rotational / local recurrent flow Ω	chart-level rotation is not anatomical feedback
HOT	none proposed	NDT does not model representation-of-representation
AST	attention-related coordinates as possible inputs	an attention proxy is not an attention schema
Landscapes / metastability	closest dynamical family	NDT adds diffusion geometry, control, and measurement contracts
Network control	possible transitions; future state space	no current B matrix or control-energy objective

Table 4: Bridges are analogies unless a shared formal object is identified. Superficial vocabulary overlap is not equivalence.

Predictive processing / free-energy principle. Precision, inference, and embodied prediction supply a neighbouring language for content and for slow bodily context (Clark, 2013; Friston, 2010; Seth, 2013). The relevant comparison is a specified generative model, not the umbrella principle. NDT's emphasis is measured, state-dependent transition structure. A fitted drift term is not variational free energy, and $r(t)$ is not precision by another name.

Integrated information. IIT identifies consciousness with irreducible intrinsic cause–effect structure, quantified as Φ (Oizumi et al., 2014; Tononi et al., 2016). Perturbational complexity is clinically important and IIT-inspired, but it is not an empirical calculation of Φ (Casali et al., 2013). NDT refuses a structural "amount of consciousness" scalar of that kind. Shared rejection of simple magnitude is not a shared ontology.

Global workspace. GNWT targets access: information becomes globally available after a competitive ignition-like transition (Baars, 1988; Dehaene & Changeux, 2011; Mashour et al., 2020). Dynamical accessibility in NDT may describe conditions that make such a transition more or less available. It does not model broadcast, report, or workspace competition. Reachable volume is not ignition.

Recurrent processing. RPT locates a threshold in recurrent sensory processing rather than in global broadcast (Lamme, 2006; 2010). NDT's antisymmetric Jacobian Ω is compatible with rotational or recurrent flow. Compatibility is not anatomy: a chart rotation does not establish feedback in sensory cortex, and recurrence is not treated as sufficient for experience.

Higher-order theories and attention schema theory. These target meta-representation and a simplified internal model of attention (Brown et al., 2019; Lau & Rosenthal, 2011; Webb & Graziano, 2015). NDT currently has no corresponding state variable. Metacognitive sensitivity and awareness attribution are neighbouring empirical programmes, not NDT objects.

11.4. Lessons that apply across theories

Two external results discipline every framework in this neighbourhood, including NDT.

The Cogitate Consortium's preregistered adversarial collaboration did not declare a winner between IIT and GNWT (Cogitate Consortium et al., 2025). Predicted posterior synchronization and predicted prefrontal offset ignition were both challenged in the specified form. The methodological lesson is prior commitment: mature theories become stronger when they state in advance what would count against them. NDT has not undergone an adversarial test of comparable independence or scale.

State discrimination is not detection of experience. Complexity measures can separate wakefulness from sedation while failing to distinguish reported dreaming from no reported experience within the sedated condition (Bajwa et al., 2025). PCI, ignition markers, and NDT reachability may all index regimes that make experience more or less likely. None is licensed as a consciousness meter by that fact. Companion coma geometry is recovery-linked capacity (Section 10), not a within-state experience assay.

12. Predictions and failure conditions

NDT is revised by layer, not as a single slogan. The predictions below are programme-level. Companion anchors constrain some of them; none is a completed adversarial result.

Prediction	What would fail it	Present constraint
P1 Rhythmic control	event structure adds nothing beyond power, autocorrelation, smooth precision, or event density	spindle-locked d_n timing; schedule specificity incomplete
P2 State vs accessibility	richer dynamics add no replicable information beyond chart location	coma, gambling, and depression move different layers
P3 Linear vs finite-amplitude	FAR collapses onto validated local-linear summaries	conceptual; not an empirical NMD export
P4 Diffusion vs drift	anisotropy / dimensionality add nothing once drift is controlled	truth-known envelopes only
P5 Thalamic access	thalamic variables add nothing beyond arousal / sensory gain	association and synthetic operators; not biology ID
P6 History dependence	$P(Y_{t+h} \mid X_t, H_t) \approx P(Y_{t+h} \mid X_t)$	measured-state prediction constraint; not landscape writing
P7 Embodied conditioning	body-state effects vanish under autonomic / arousal controls	timing support; spindle-geometry efficacy rejected

Table 5: Falsifiable predictions. Success or failure of one row revises that layer. It does not automatically ratify or refute the modelling class.

A stronger NDT discriminant would jointly require event specificity, schedule specificity against stationary and smooth controls, incremental value beyond power and generic complexity, and failure under phase-shuffle, time-shift, wrong-event, and power-matched nulls. Current spindle evidence supports a timing-locked chart change. It does not yet satisfy the full discriminant.

Hierarchical failure

Falsification of one subsystem does not falsify dynamical accessibility as a scientific object. TRN failure would drop a biological implementation. RVC failure would strongly constrain rhythmic scheduling. MNPS failure would retire that chart. Operator-recovery failure on a given recording, already observed on DANDI 000458, retires that estimator on that chart. None of those results implies that neural states have no structured futures.

NDT should be substantially narrowed if, across reasonable alternative charts and nulls: rhythmic scheduling adds no held-out value; local-dynamics estimators repeatedly fail truth-known or residual-quality tests; diffusion, reachability, or FAR reduce to redundant versions of simpler statistics; history adds no conditional information; and apparent clinical or altered-state effects are exhausted by power, movement, arousal, or preprocessing artifacts. The constitutive reading that rhythmic reconstruction **is** experience remains a labeled speculative extension. It currently adds no discriminating empirical prediction beyond the dynamical model.

13. Discussion

13.1. From denoising to accessibility

NDT began as a theory of rhythmic reconstruction: noise, coherence, a learned potential, a variance schedule, and a geometric state chart. That identity is not discarded. It is no longer the centre. The scientifically load-bearing object is now the local dynamical neighbourhood of a neural state — what the system tends to do, how fluctuations spread, which nearby futures remain open, and which processes change those possibilities.

The original emphasis was therefore

$$\text{noise} \rightarrow \text{coherence} \rightarrow \text{denoising} \rightarrow \text{rhythmic schedule} \rightarrow \text{geometric state}.$$

The revised emphasis is

state → local law → perturbation → possible futures → control of those futures.

That is a change of question, not a relabeling of MNPS. Location remains necessary. It is incomplete as an account of organization. Companion evidence already forces the shift: spindle events reconfigure a stratified chart without licensing causal denoising; early post-arrest EEG separates more clearly on reachable futures and local deformation than on static spread; meso-to-macro contact does not recover a local operator (Section 10). The theory follows those constraints rather than protecting an earlier slogan.

13.2. What is retained, promoted, and separated

Retained from the original programme are the exploration–stabilization trade-off, learned landscapes as a structured component of drift, rhythmic control as a distinctive temporal hypothesis, geometric state representations, metastability as a neighbouring dynamical regime, and explicit layered falsification.

Promoted to first-class objects are the diffusion tensor, the local Jacobian and its stability/reactivity summaries, short-horizon reachability, committor geometry, finite-amplitude resilience, and attractor/basin structure — each as an answer to a different dynamical question, not as interchangeable complexity scores (Table 1).

Separated from the theory proper is NeuralManifoldDynamics. Charts and estimators are versioned empirical realizations. They can fail without making dynamical accessibility meaningless, and they can succeed as measurements without identifying the latent law. Generalized from a scalar schedule is accessibility control: rhythm may act on $\sigma(t)$, but context, history, embodiment, and metabolism may act on drift, mobility, landscape, or transition gates (Section 9). RVC remains the minimal special case.

Demoted, relative to earlier drafts, are a consciousness scalar, a universal clinical fingerprint, Mapper/topology-as-theory, and any reading in which MNPS location **is** the NDT claim.

13.3. Why this may matter

A conscious brain state may not be distinguished purely by complexity, entropy, synchrony, occupancy, or global activity. Candidate regimes may also differ in **what the system remains capable of becoming next**. That is a hypothesis about the organization of neural states associated with wakefulness, sleep, anesthesia, task engagement, and pathology. It is not an identity claim about experience, not a meter of awareness, and not a solution to the hard problem.

The hypothesis is useful only if the layers dissociate. If reachability, local flow, and diffusion were redundant encodings of location or power, the revised centre would collapse. The present empirical map is not a validation of the chain. It is evidence that the collapse has not already occurred in every tested regime, and that different conditions move different layers (Table 3). Consciousness research already contains neighbouring theories of content, access, recurrence, and intrinsic causal structure (Section 11). NDT's contribution, if it holds, is a measurement surface for dynamical capacity and its control — a surface on which those theories can make distinct predictions, not a replacement for them.

The next informative experiment is therefore not another broad disease contrast. It is a design that separates global state, responsiveness, reportability, and momentary experience, and that asks whether rhythm-locked or context-locked changes in local flow, diffusion, and short-horizon reachability explain future trajectories beyond power, generic complexity, and stationary dynamics (Section 12). A positive result would still support a neural mechanism, not the claim that reconstruction is consciousness.

14. Limitations

No dataset currently validates the NDT chain end to end. Many nominated objects remain estimator-sensitive: MNPS is operational and chart-dependent; drift and diffusion are identifiable only under explicit assumptions; local linearization fails in important empirical regimes, including the EEG chart tested on DANDI 000458; reachability depends on horizon and noise model; committor estimation requires meaningful destinations that current measurement contracts do not emit; and finite-amplitude resilience depends on perturbation family and amplitude scale.

Biological bridges remain uneven. Rhythmic variance control is a distinctive prediction, not a demonstrated executor. Thalamic access has synthetic operator existence and associative biology, not circuit identification. History-dependent landscape writing is open; access on a frozen landscape is a computational existence result, not scored hippocampal mechanism. Embodiment and metabolism are contextual and feasibility constraints, not consciousness coordinates. Dendritic slow-branch accessibility is a simulator discriminator and a speculative micro substrate.

Empirical companions are bounded. Sleep, coma, and meso-to-macro analyses constrain selected layers in selected cohorts. Clinical fingerprints are not biomarkers. Treatment-limitation and sedation confounds remain unresolved where they were unavailable. NDT has not undergone an independent adversarial test comparable to Cogitate. Apparent geometric effects must still survive alternative charts, power-matched and temporally shuffled nulls, and standard spectral, connectivity, and complexity baselines.

These limits are scope, not temporary embarrassment. NDT is a framework for neural dynamical organization associated with conscious-state regimes. It does not explain why those dynamics are accompanied by experience.

15. Conclusion

Noetic Diffusion Theory proposes that neural-state organization is incompletely described by state position alone. A neural state also carries a local dynamical structure: deterministic tendencies, stochastic directions, transient sensitivities, reachable futures, basin commitments, and finite-amplitude resilience. Those structures are themselves modulated by rhythm, context, history, physiology, and candidate access-control systems. MNPS supplies one measurable chart through which this organization can be studied. NeuralManifoldDynamics supplies one implementation rather than the definition of the theory. Rhythmic Variance Control supplies the minimal control case rather than the whole mechanism.

The resulting framework predicts structured dissociations among location, stability, diffusion, accessibility, commitment, and resilience, and treats those dissociations as empirically testable correlates of changing neural regimes — not as a scalar of consciousness, and not as a complete explanation of experience. What a neural state **is**, scientifically, includes what it can still become.

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Data Availability Statement

This theoretical manuscript will link to public datasets, simulator code, analysis contracts, and permanent archives supporting each bounded companion claim. Sleep, I-CARE, and DANDI 000458 sources used in the empirical section are cited there.

Competing Interests

The author declares no competing interests, subject to final confirmation at submission.